

### SCHEMA DESIGN FOR THE PROJECT:

1. Identifying Entities
2. Defining Entities, Attributes and Relationships
3. ER Diagram
4. Relationships (1-to-1, 1-to-many..)

## 1. Identifying Entities & Attributes

- Users: **id**, name, email, role (contributor / moderator / admin), created\_at, updated\_at

| Column Name       | Data Type    | Description                                                                |
|-------------------|--------------|----------------------------------------------------------------------------|
| <b>id</b>         | Integer (PK) | Unique identifier for the user.                                            |
| <b>name</b>       | String       | Name of the user.                                                          |
| <b>email</b>      | String       | Email of the user.                                                         |
| <b>role</b>       | Enum         | Role of the user ( <b>contributor</b> , <b>moderator</b> , <b>admin</b> ). |
| <b>created_at</b> | Timestamp    | Timestamp of record creation.                                              |
| <b>updated_at</b> | Timestamp    | Timestamp of the last record update.                                       |

- Query Table:

| Column Name | Data Type    | Description                                     |
|-------------|--------------|-------------------------------------------------|
| id          | Integer (PK) | Unique identifier for the message.              |
| user_id     | Integer (FK) | ID of the user who created the message.         |
| content     | Text         | The content of the message.                     |
| tags        | Array        | Tags or mentions associated with the message.   |
| created_at  | Timestamp    | Timestamp of when the message was created.      |
| updated_at  | Timestamp    | Timestamp of when the message was last updated. |
| likes       | Integer      | Number of upvotes for a message                 |
| status      | Integer      | Open, closed, in progress                       |
| flagged     | boolean      | Flagged query                                   |

- Responses Table:

| Column Name | Data Type    | Description                                                  |
|-------------|--------------|--------------------------------------------------------------|
| id          | Integer (PK) | Unique identifier for the message.                           |
| user_id     | Integer (FK) | ID of the user who created the message.                      |
| content     | Text         | The content of the message.                                  |
| query_id    | Integer (FK) | ID of the parent message (NULL if it's an original message). |
| tags        | Array        | Tags or mentions associated with the message.                |
| created_at  | Timestamp    | Timestamp of when the message was created.                   |
| updated_at  | Timestamp    | Timestamp of when the message was last updated.              |
| likes       | Integer      | Number of upvotes for a message                              |
| upvotes     | Integer      | Favourable                                                   |
| downvotes   | Integer      | Unfavourable                                                 |
| approval    | boolean      | Response approved by user who put query                      |
| flagged     | boolean      | Flagged query                                                |

- Moderation Logs:

| Column Name           | Data Type    | Description                                                                           |
|-----------------------|--------------|---------------------------------------------------------------------------------------|
| <code>id</code>       | Integer (PK) | Unique identifier for the moderation log.                                             |
| <code>query_id</code> | Integer (FK) | Status of the query                                                                   |
| <code>action</code>   | Enum         | Action taken ( <code>approved</code> , <code>flagged</code> , <code>deleted</code> ). |

- Tags  
`id`, `name`, `created_at`

| Column Name             | Data Type    | Description                    |
|-------------------------|--------------|--------------------------------|
| <code>id</code>         | Integer (PK) | Unique identifier for the tag. |
| <code>name</code>       | String       | Name of the tag.               |
| <code>created_at</code> | Timestamp    | Timestamp of record creation.  |

## 2. Defining Entities

- Users: individuals who will interact with the platform, categorized as - Contributor, Moderator, Admin
- Issues: tracks the problems raised by contributors within the community.
- Responses: answers, conversation threads or comments provided to address the issue
- Moderation Logs: to record the actions initiated by moderators.
- Leaderboards: tracking the contributions made in the community, to encourage participation.
- Tags: Used to categorize the issues for better understanding.

## 3. Relationships

- User → Queries : One to many
- Queries → Responses : one to many

- Responses → tags: one to many
- Queries → Tags: one to many
- Query → moderation\_log: One to one

## FLOW OF THE APPLICATION:

1. User (Contributor)
  - a. Can create multiple queries - opening, closing, approving responses
  - b. Can respond to others' queries
    - Responses can be flagged if inappropriate
    - Flagged responses will be visible to Admin - to take action
    - They can be flagged, upvoted or downvoted by anyone.
2. Conversation organised with tags
3. Flagged queries/responses can be deleted by the users.
4. Notifications - for new queries / once a query is closed.

# ER Diagram:

